

Chapter 1

Web Introduction & HTML for Data

Theory & Concepts

This theoretical session will ensure students have a fundamental understanding of the web ecosystem, HTML's role as the backbone of content, and the importance of code version control.

1.1 Web Fundamentals and Web-Based Accounting Applications:

- **What are the Internet and WWW?** A brief explanation of the global network and hypertext information system.
- **HTTP (Hypertext Transfer Protocol) & HTTPS:** How browsers (clients) and servers communicate. The importance of HTTPS for data security (relevant for financial transactions).
- **URL (Uniform Resource Locator):** URL structure (protocol, domain, path, parameters) and its significance in accessing data.
- **Web Browser as a Client:** The main functions of a browser (HTML parsing, CSS rendering, JavaScript execution).
- **Roles of Front-end vs. Back-end Development:** Illustration with examples of accounting applications:
 - **Front-end:** Display of transaction input forms, financial report dashboards, data tables.
 - **Back-end:** Calculation processes, transaction data storage, user authentication, report access authorization.
- **Concepts of Static vs. Dynamic Web:** HTML as the static base, the need for server-side languages (PHP) for dynamic content like changing reports.

1.2 Anatomy of an HTML Document (Hypertext Markup Language):

- **Purpose of HTML:** The standard markup language for creating the structure and content of web pages.
- **Required HTML Document Structure:**
 - `<!DOCTYPE html>`: Document type declaration (important for browser rendering).
 - `<html>`: The root element of every HTML document. lang attribute (e.g., lang="en") for accessibility and SEO.
 - `<head>`: Section for page metadata (not directly visible in the browser).
 - `<meta charset="UTF-8">`: Character encoding specification (important for supporting special characters, e.g., currency symbols).
 - `<meta name="viewport" content="width=device-width, initial-scale=1.0">`: Viewport settings for initial responsiveness.
 - `<title>`: The page title that appears in the browser tab.
 - `<link rel="stylesheet" href="style.css">`: How to link external CSS files (to be

discussed in Chapter 4).

- `<script src="script.js"></script>`: How to link external JavaScript files (to be discussed in Chapter 7).
- `<body>`: The main part of the web page that will be displayed in the browser (visual content).

1.3 Basic HTML Elements for Content and Initial Structure:

- **Headings (`<h1>` to `<h6>`):** Hierarchy of titles for organizing content (e.g., `<h1>Financial Report</h1>`, `<h2>Revenue</h2>`). Important for SEO and semantic structure.
- **Paragraph (`<p>`):** For blocks of plain text (e.g., description of accounting policies).
- **Links (`<a>` - **Anchor Tag**):**
 - Internal links (between pages within the project).
 - External links (`target="_blank"` to open in a new tab).
 - Email links (`mailto:`) and phone links (`tel:`).
- **Images (`<img>`):**
 - `src` attribute (image source: relative path or URL).
 - `alt` attribute (alternative text): Very important for accessibility (screen readers) and SEO if the image fails to load or for search bots.
 - `width` and `height` attributes (CSS is recommended for styling).
- **Line Break (`<br>`):** To force a new line within text.
- **Horizontal Rule (`<hr>`):** For a horizontal line as a content separator.
- **Lists (`<ul>`, `<ol>`, `<li>`):** For displaying bulleted or numbered lists (e.g., chart of accounts, list of assets).
 - `<ul>` (Unordered List): Bulleted list.
 - `<ol>` (Ordered List): Numbered list.
 - `<li>` (List Item): An item in a list.
- **Basic Text Formatting Tags:** `<strong>` (bold, important), `<em>` (italic, emphasis), `<u>` (underline - use with caution).

1.4 Introduction to Version Control Systems (VCS) & Git (Fundamentals):

- **What is VCS and Why Do We Need It?**
 - Records every change to the code (version history).
 - Enables team collaboration without overwriting each other's work (relevant for group projects).
 - Ability to revert code to a previous version (if errors occur).
 - Branch management (feature development branches).
- **Git as the Most Popular VCS:** Distributed system, fast, flexible.
- **Basic Git Concepts:**
 - **Repository (Repo):** A Git project storage location. Can be local or remote.
 - **Working Directory:** The project folder where we work.
 - **Staging Area (Index):** A temporary area to select changes to be saved.

- **Commit:** A snapshot of changes saved to the repository's history.
- **Essential Git Commands (Initial):**
 - `git --version`: Checks Git installation.
 - `git init`: Initializes a new Git repository in the local folder.
 - `git status`: Shows the status of files (changes not yet added, added, not yet committed).
 - `git add <file_name>` or `git add .`: Adds changes to the staging area.
 - `git commit -m "Commit message"`: Saves changes from the staging area to history with a descriptive message.
 - `git log`: Views commit history.

Practical Exercises

This section will heavily focus on hands-on practice, ensuring each student gains maximum coding experience, with themes relevant to Accounting Information Systems.

Environment Setup & Class Exercises

1. Essential Tools Installation (Complete Guide for Windows):

1.1.1. Web Browser Installation (Google Chrome/Mozilla Firefox):

- **Purpose:** Used to view coding results and for debugging with Developer Tools.
- **Guide:**
 1. Open an existing browser (e.g., Microsoft Edge).
 2. Visit <https://www.google.com/chrome/> for Google Chrome or <https://www.mozilla.org/firefox/> for Mozilla Firefox.
 3. Click the "Download" button.
 4. Once the .exe file is downloaded, double-click it and follow the on-screen installation instructions.
 5. Ensure you can successfully open this new browser.

1.1.2. Visual Studio Code (VS Code) Installation:

- **Purpose:** As the primary text editor for writing HTML, CSS, JavaScript, and PHP code.
- **Guide:**
 1. Visit <https://code.visualstudio.com/download>.
 2. Select the download for **Windows (User Installer)**, matching your system's architecture (usually x64).
 3. Once the .exe file is downloaded, double-click it.
 4. Follow the installation steps:
 - Select "I accept the agreement".
 - Click "Next".
 - Choose the installation location (keep default if no special needs).
 - In the "Select Additional Tasks" window, **it is highly recommended** to check all options:

- Create a desktop icon
- Add "Open with Code" action to Windows Explorer file context menu
- Add "Open with Code" action to Windows Explorer directory context menu
- Register Code as an editor for supported file types
- Add to PATH (requires shell restart) - **Important for using Git in the terminal.**
- Click "Next", then "Install".

5. After installation is complete, open VS Code.

1.1.3. Git Installation for Windows:

- **Purpose:** For managing your code versions and collaboration.
- **Guide:**
 1. Visit <https://git-scm.com/download/win>.
 2. Download the Git installer (usually the 64-bit version for Windows Setup).
 3. Once the .exe file is downloaded, double-click it.
 4. Follow the installation steps:
 - Click "Next" on the "GNU General Public License".
 - Choose the installation location (keep default).
 - Select components (keep default if unsure, or check "Git Bash Here", "Git GUI Here" and "Associate .git* configuration files with the default text editor").
 - In "Adjusting your PATH environment", select **"Git from the command line and also from 3rd-party software"** (This option is crucial for accessing Git from CMD/PowerShell/VS Code terminal).
 - Select default options for the rest until completion.
 - Click "Install".
 5. After installation is complete, verify Git by opening **Command Prompt (CMD)** or **Git Bash** and typing: `git --version`. Ensure the version number appears.

2. Creating the Main Project Folder Structure:

- Each student creates a main folder in an easily accessible location, e.g., `C:\Users\YourName\Documents\` or `D:\MyProjects\`. Name the main folder: `basic_programming_ais_YourNIM/`. Example: `basic_programming_ais_A123456789/`.
- Inside this main folder, create a `week_1/` folder. This will be the working directory for Chapter 1.

Per-Theory Sub-Chapter Exercises

These exercises are designed to immediately apply each theoretical point taught, ensuring a deep understanding before moving on to the individual practical assignment.

Exercise 1.2.1: Basic HTML Structure & Essential Metadata.

- Inside the `week_1/` folder, create a new HTML file named `basic_html.html`.

- Type ! then press Tab in VS Code to auto-generate the basic HTML structure.
- Modify the <title> tag to: Basic HTML Learning for AIS.
- Ensure <meta charset="UTF-8"> and <meta name="viewport" content="width=device-width, initial-scale=1.0"> are present within the <head>.
- Inside the <body>, add <h1>This is My First Page for Accounting Information Systems Course</h1>
- Save the file (Ctrl+S) and open it in your browser. Observe the browser tab title.

Exercise 1.3.1: Using Headings & Paragraphs for Accounting Information.

- In the same basic_html.html file, add after <h1>:

```
<h2>Simple Income Statement</h2>
<h3>For the Period January 2025</h3>
<p>This report summarizes the company's revenues and expenses for a one-month period.</p>
<p>All figures are presented in thousands of Indonesian Rupiah (IDR), and are unaudited. This information is illustrative for learning purposes.</p>
```

Save and view the results in the browser. Note the difference in heading sizes and how paragraphs separate blocks of text.

Exercise 1.3.2: Creating an Account List and Relevant Links.

- In the basic_html.html file, add after the last paragraph:

```
<h2>Main Account List (Illustrative)</h2>
<ul>
  <li>Cash</li>
  <li>Accounts Receivable</li>
  <li>Inventory</li>
  <li>Land, Buildings, and Equipment</li>
  <li>Accounts Payable</li>
  <li>Sales Revenue</li>
  <li>Salaries Expense</li>
  <li>Share Capital</li>
</ul>

<h2>Additional Resources</h2>
<p>Visit the <a href="https://www.ojk.go.id/" target="_blank">Financial Services Authority (OJK) Website</a> for financial regulatory information in Indonesia.</p>
<p>If you have any questions about the material, please <a href="mailto:ahmad.wafa@polines.ac.id">Email Your Lecturer</a>.</p>
<p>You can also see examples of financial statements on the <a href="https://www.idx.co.id/" target="_blank">Indonesia Stock Exchange (IDX) Website</a>.</p>
```

- Save and test the created links. Notice target="_blank" will open the link in a new tab.

Exercise 1.3.3: Inserting Illustrative Images & Basic Text Formatting.

- Create a folder `assets/images/` inside `week_1/`. Download a small relevant image (e.g., a chart icon, generic accounting logo, or calculator image) and save it in this folder. You can use a placeholder URL if you have trouble finding an image:
`https://placeholder.co/150x100?text=Chart`
- In the `basic_html.html` file, add at the top of the `<body>`, below `<h1>`:

```

<hr> <!-- Visual separator line -->
<p>This is a <strong>summary report</strong> that is
<em>important</em> for Accounting Information Systems students to
understand.</p>
```

- Ensure the image appears. Try changing the `src` to an incorrect file name and observe how the alt text is displayed. Also, observe the effects of `<strong>` and `<em>`.
- **Exercise Git 1.4: Initializing & Committing the Project.**
 - Open the integrated terminal in VS Code (Terminal > New Terminal) in your project's root folder (`basic_programming_ais_YourNIM/`).
 - Run the Git commands:

```
git init
git add week_1/basic_html.html week_1/assets/images/
git commit -m "Class Exercise Commit: Basic HTML structure, headings,
links, images, lists"
```
 - Use `git status` and `git log` to verify that changes have been successfully saved in Git history.

Individual Practical Assignment

Students will complete this assignment individually as part of their weekly homework. This assignment will test their comprehensive understanding of all concepts and elements taught in Chapter 1. Students are encouraged to be creative with the content, as long as all mandatory requirements are met.

1. Assignment 1.1: Web Page "Profile of a Tax Consultant & Tax Services"

- Create a new folder named `tax_consultant_profile/` inside the `week_1/` folder.
- Inside `tax_consultant_profile/`, create a new HTML file named `index.html`.
- Also, create a folder `tax_consultant_profile/assets/images/`. Download 2-3 relevant placeholder images related to tax/consulting (e.g., a fictional consultant logo, office building image, or an illustration of someone calculating tax) and save them in this folder. You can use image URLs from `https://picsum.photos/` or `https://via.placeholder.com/` if local images are unavailable.

- **Task:** Design a web page that provides an overview of a "fictional Tax Consulting Office" and details the tax services offered.
- **Mandatory Requirements (All must be included):**
 - Use the complete HTML structure (<!DOCTYPE html>, <html>, <head> with meta, title, <body>).
 - Use at least 3 levels of **headings** (<h1>, <h2>, <h3>) for the main title (Office Name), sub-sections (e.g., "Profile", "Services", "Contact"), and important points (e.g., "Our Advantages").
 - Include at least 3 different **paragraphs** for office description, mission, or vision.
 - Involve at least 2 **images** () with descriptive alt attributes and correct paths (relative path from index.html).
 - Include at least 2 **external links** (<a> with target="_blank") to relevant tax websites (e.g., Directorate General of Taxes (DJP) website, or tax news sites).
 - Include at least 1 **email link** (mailto:).
 - Include at least 1 **phone link** (tel:).
 - Use at least one **unordered list** () for the list of main services (e.g., Corporate Income Tax, VAT, Annual Tax Return) and one **ordered list** () to explain the process/steps in a service (e.g., 3 consultation steps).
 - Use basic text formatting tags like (to emphasize keywords like "Tax", "Efficiency") and (for emphasis).
 - Add at least two **horizontal rules** (<hr>) as separators between main sections of the page.
 - Ensure all elements used have content relevant to the Tax Consultant theme.
- **Save Git Changes:** After all HTML files and images are ready, open the terminal in your project's root folder (basic_programming_ais_YourNIM/) and run the commands:

```
git add week_1/tax_consultant_profile/
git commit -m "Completed Assignment 1.1: Tax Consultant Profile and Services"
```

Submission Method

Students will create a **PDF document** containing:

1. **HTML Code** from week_1/tax_consultant_profile/index.html (Assignment 1.1). Ensure the code is neat, properly indented, and easy to read.
2. **Screenshots of the index.html page in the browser.** Ensure all elements (images, links, lists, text formatting, headings) are clearly visible and the layout is appropriate. Also include screenshots when clicking on some links (e.g., external links and email/phone links) to demonstrate functionality.
3. **Screenshot of the terminal/CMD** showing the output of the git log command after the last commit for Assignment 1.1. This is to verify that the commit has been made.
4. The PDF should be printed and submitted to the lecturer for signature.